

DATA ANALYTICS

Unsupervised Anomaly Detection
in Mars HiRISE Orbital Imagery

Technical Solution & Results Report

Challenge	Data Analytics
Dataset	10,422 single-channel grayscale Mars HiRISE crops
Final model	V3 convolutional autoencoder + Isolation Forest
Latent representation	256-dimensional learned vector
Threshold	0.626565 ($\mu + 3\sigma$)
Candidates	206 (1.98%)
Top-1 result	sample_06029.jpg — novelty score 0.742821

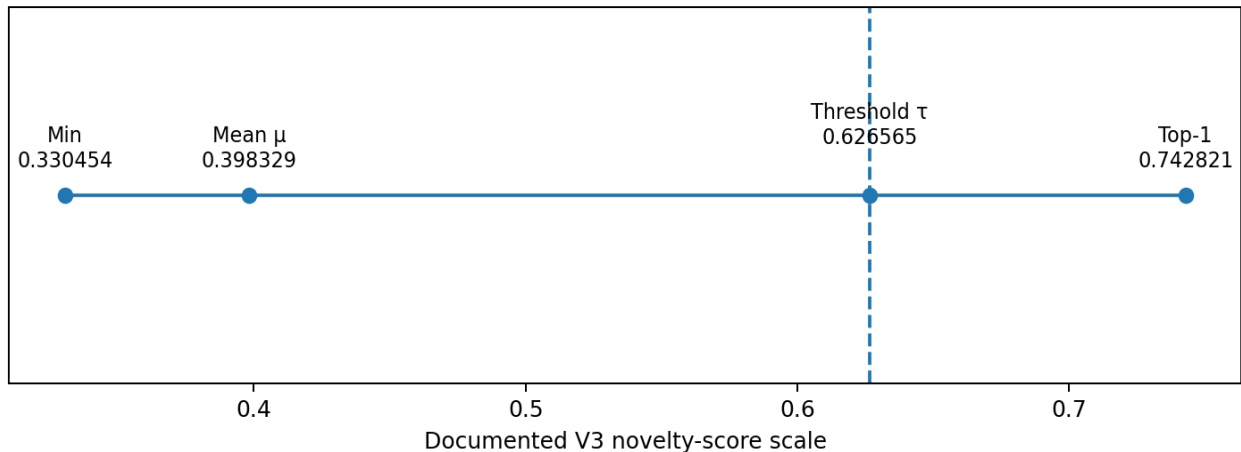
1. Executive Summary

This project develops an end-to-end unsupervised anomaly detection pipeline for Mars HiRISE orbital imagery. The objective is to learn a compact representation of predominantly normal Martian terrain from unlabeled grayscale crops, then identify samples that are statistically isolated in the learned feature space.

The final system uses Autoencoder Version 3 (V3), a four-stage convolutional encoder-decoder with symmetric skip connections, a 256-dimensional fully connected bottleneck, and a combined mean-squared-error (MSE) plus gradient-aware structural loss. A 300-tree Isolation Forest scores the resulting latent vectors.

Using the documented empirical rule $\tau = \mu + 3\sigma$, the final threshold is 0.626565. This produces 206 threshold-qualified candidates, representing 1.98% of the 10,422-image dataset. The highest novelty score is 0.742821 for sample_06029.jpg.

Figure 1. Statistical threshold relationship (schematic; based on reported summary statistics)



The threshold graphic above is a schematic of the documented summary statistics, not a reconstructed empirical histogram.

FINAL RESULT
206 anomaly candidates • 10,216 normal images • 1.98% candidate rate
Top-1: sample_06029.jpg • Novelty: 0.742821 • Threshold: 0.626565

2. Problem Statement & Objectives

HiRISE observations produce large volumes of orbital imagery. Manual inspection of tens of thousands of crops for rare or scientifically important features is impractical, while uncharted planetary surfaces do not provide reliable ground-truth anomaly labels.

- Unsupervised representation learning: learn normal Martian surface morphology without labels.
- Latent compression: map each 227×227 grayscale crop into a 256-dimensional bottleneck.
- Novelty detection: use Isolation Forest to quantify statistical isolation in latent space.
- Statistical thresholding: derive an anomaly boundary from the empirical score distribution.
- Interpretability: localize reconstruction differences using pixel-wise error maps.
- Metadata analysis: compare detected candidates with acquisition parameters.

No-pretraining principle: the documented pipeline uses no pretrained backbone, pretrained weights, transfer learning, or external feature extractor; representations are learned strictly from the provided dataset.

3. Dataset & Preprocessing

Item	Specification
Total images	10,422

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Image format	JPEG (.jpg), single-channel grayscale, 8-bit L-mode
Crop dimensions	227 × 227 pixels
Training set	8,337 images (80.0%)
Validation set	2,085 images (20.0%)
Split seed	random_state = 42
Normalization	uint8 → float32 by division by 255; channel dimension added
Metadata	crop_metadata_index.csv + source_image_metadata.csv; 172 source observations
Raw-pixel sanity check	Min 0; Max 255; Mean 118.439095; Std 46.071037; Median 123

The documented preprocessing preserves spatial fidelity: no synthetic resizing, cropping, padding, or artifact removal is applied.

4. End-to-End Pipeline

Figure 2. End-to-end pipeline architecture

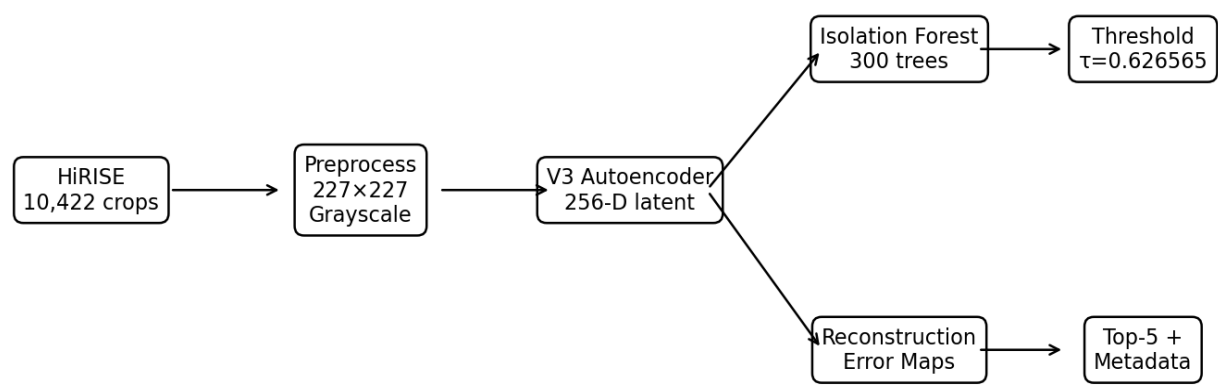


Figure 2. End-to-end processing flow based on the documented implementation.

The pipeline is sequential: normalized images are encoded into a learned latent representation; the full latent matrix is scored by Isolation Forest; a statistical threshold selects candidates; reconstruction error maps and acquisition metadata provide additional interpretation.

5. Final Model Architecture — Autoencoder V3

V3 uses a four-stage convolutional encoder-decoder with a fully connected 256-D bottleneck. Symmetric skip connections concatenate intermediate encoder activations with decoder representations, preserving multi-scale spatial information.

Stage	Layer / operation	Output
Input	Normalized grayscale tensor	1 × 227 × 227
Encoder 1	Conv2d 1→32, k=3, s=2, p=1 + ReLU	32 × 114 × 114
Encoder 2	Conv2d 32→64, k=3, s=2, p=1 + ReLU	64 × 57 × 57
Encoder 3	Conv2d 64→128, k=3, s=2, p=1 + ReLU	128 × 29 × 29
Encoder 4	Conv2d 128→256, k=3, s=2, p=1 + ReLU	256 × 15 × 15
Bottleneck	Flatten 57,600 → Linear → 256	256-D latent vector

Stage	Layer / operation	Output
Decoder	Linear 256 → 57,600 → reshape	256 × 15 × 15
Skip blocks	ConvTranspose2d + encoder-feature concatenation	Multi-scale detail recovery
Output	ConvTranspose2d 64→1 + Sigmoid	1 × 227 × 227

Total trainable parameters: 30,416,929. Training is strictly from scratch.

6. Loss Function & Model Evolution

V1 and V2 use pure MSE. V3 retains the V2 skip-connected topology and augments the objective with horizontal and vertical finite-difference gradient terms. The documented V3 objective is MSE + 0.1 × gradient-aware structural loss.

Version	Topology / loss	Symptom → Diagnosis → Fix → Result
V1 — Baseline	4-stage ConvAE; pure MSE; no skip connections	Symptom: broad terrain reconstructed but fine ridges, small craters and high-frequency textures were smoothed. Diagnosis: compressed spatial detail had to be regenerated after the bottleneck. Fix: establish baseline 4-stage convolutional autoencoder. Result: best Val MSE 0.002040 at Epoch 19.
V2 — Improved	4-stage ConvAE + symmetric skips; pure MSE	Symptom: localized detail was lost through the bottleneck. Diagnosis: intermediate encoder maps were discarded. Fix: multi-scale symmetric skip connections. Result: documented 98.16% validation-error reduction over V1 and sharper boundaries.
V3 — Final	V2 topology + MSE + 0.1 × gradient-aware loss	Symptom: visually sharp reconstructions still lacked an explicit gradient-sensitive objective. Diagnosis: pixel MSE does not directly emphasize boundary transitions/orientation. Fix: finite-difference gradient terms. Result: enhanced edge/ridge fidelity and interpretability.

Selection rationale: V2 achieved the lowest pure validation MSE, but V3 was selected because its combined objective explicitly incorporates local gradient and structural information valuable for morphological error interpretation.

7. Latent-Space Visualization

t-SNE was applied to the 2,085 validation latent vectors using `n_components=2`, `perplexity=30`, `learning_rate='auto'`, `init='pca'`, and `random_state=42`.

- The 2-D projection is a qualitative diagnostic of representation structure.
- A continuous central population represents typical background Martian terrain.
- Several separated groupings are visible.
- The groupings are not treated as confirmed geological classes or anomaly labels.
- Final anomaly identification is performed independently in the full 256-D latent space using Isolation Forest.

Figure availability note: the supplied project document describes the t-SNE observations and parameters but does not contain the original rendered t-SNE image. No synthetic t-SNE plot is presented here.

8. Isolation Forest Novelty Detection

Parameter	Documented value
Estimator	Isolation Forest
Trees	300
max_samples	auto
contamination	auto; decoupled from final threshold decision
random_state	42
n_jobs	-1
Native scoring	score_samples gives lower values to more isolated samples
Project novelty score	Native score is negated so higher values indicate greater anomaly
Score range	0.330454 to 0.742821
Mean μ	0.398329

Parameter	Documented value
Standard deviation σ	0.076079
Threshold τ	$\mu + 3\sigma = 0.626565$

The final threshold is intentionally empirical rather than an arbitrary contamination percentage. Scores above τ are treated as threshold-qualified anomaly candidates.

9. Final Anomaly Results

Rank	Filename	Novelty score	Decision
1	sample_06029.jpg	0.742821	Exceeds τ
2	sample_04537.jpg	0.742797	Exceeds τ
3	sample_03547.jpg	0.739186	Exceeds τ
4	sample_09253.jpg	0.738691	Exceeds τ
5	sample_08233.jpg	0.737548	Exceeds τ

206 candidates	10,216 normal images	1.98% of dataset
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Final answer: the highest-scoring candidate is **sample_06029.jpg** with novelty score **0.742821**; all five Top-5 candidates exceed $\tau = 0.626565$.

10. Reconstruction Interpretability & Geological Hypotheses

Each candidate is passed through V3 to obtain a reconstruction. Pixel-wise squared error $E(i,j) = (x(i,j) - \hat{x}(i,j))^2$ is used to localize regions where the learned normal-terrain model struggles to reproduce the input.

Rank / image	Observed morphology	Error evidence / hypothesis
1 — sample_06029.jpg	Multiple elongated, approximately parallel high-contrast ridge/trough-like structures.	Differences concentrate along linear structures. Possible layered, ridged or trough-like terrain; illumination/shadow geometry may contribute.
2 — sample_04537.jpg	Irregular high-contrast structure near the right side.	Mismatch concentrates around the structure. Possible unusual local morphology or texture transition; black boundary is treated as an image border.
3 — sample_03547.jpg	Elongated and irregular high-contrast structures in smoother terrain.	Highest mean and maximum pixel error among Top-5. Possible ridge-like, eroded, or elevated/depressed structures; acquisition/illumination effects remain possible.
4 — sample_09253.jpg	Compact dark-and-bright oval/circular feature with pronounced boundary.	Error concentrates around the circular feature. Possible small depression, pit-like feature, or shadowed surface object.
5 — sample_08233.jpg	Elongated bright ridge-like structures in textured terrain.	Localized differences occur around bright elongated structures. Possible ridge-like/layered morphology; solar illumination may enhance relief.

Interpretation caution: these are observational hypotheses, not confirmed geological classifications or discoveries.

11. Metadata & Acquisition Analysis

The 206 detected candidates originate from 101 unique source images. The documented comparison indicates that candidates are distributed across multiple parent observations rather than being concentrated in one corrupted strip.

Variable	Anomaly subset	Full dataset	Observation
N-spring	42.23%	40.20%	Closely matched
N-summer	27.67%	29.07%	Closely matched
N-autumn	19.42%	20.93%	Closely matched
N-winter	10.68%	9.80%	Closely matched
Resolution 0.25 m/px	51.94%	51.10%	Distributed across standard resolutions

Variable	Anomaly subset	Full dataset	Observation
Resolution 0.50 m/px	46.60%	47.58%	Distributed across standard resolutions
Resolution 1.00 m/px	1.46%	1.31%	Distributed across standard resolutions
Sun angle	32.10°–85.68°	32.10°–85.68°	Identical documented range

The documented metadata analysis does not show concentration in a particular season, solar-illumination range, or single-strip imaging artifact. Nevertheless, illumination and image-boundary effects remain recognized limitations.

12. Competition Submission Requirement Coverage

Requirement	Coverage in project
Well-documented .ipynb	Notebook is documented as NSSC_2026_Mars_HiRISE_Anomaly_Detection.ipynb and contains the implementation/outputs.
PDF report	This report provides structured explanations, tables, documented results, model evolution, interpretability and limitations.
Plots / figures	Two figures are included: end-to-end pipeline architecture and a clearly labelled statistical-threshold schematic. The supplied document did not include the original rendered t-SNE/heatmap images.
Markdown model evolution	V1 → V2 → V3 evolution is documented using the required symptom → diagnosis → fix structure.
GitHub	Repository structure documented as nssc-data-analytics/ with notebook and README.md.
Reproducibility	Seed 42, deterministic checkpoint restoration, shuffle=False latent extraction, and automated assertions are documented.

13. Technologies & Reproducibility

Area	Technology / configuration
Language	Python
Deep learning	PyTorch
Novelty detection	Scikit-learn IsolationForest
Latent visualization	Scikit-learn TSNE
Scientific computing	NumPy, Pandas, SciPy
Image processing	Pillow
Visualization	Matplotlib and Seaborn
Hardware	NVIDIA CUDA; Tesla T4 verified in notebook
Random seed	42
Checkpointing	Best validation-loss state saved/restored; Epoch 19 documented for V1/V2/V3
Data alignment	shuffle=False, num_workers=0
Assertions	Latent dimensions, score lengths, threshold and Top-5 criteria

14. Limitations

- **Unsupervised paradigm:** without ground-truth labels, detections are statistical outliers, not confirmed geological phenomena.
- **Image boundary effects:** black borders or detector edge margins can produce elevated reconstruction error.
- **Illumination and shadows:** extreme solar incidence and topography can generate strong gradients and contrast.
- **Single-channel imagery:** grayscale analysis lacks multispectral/compositional context.
- **Interpretation uncertainty:** morphology-based hypotheses require scientific validation beyond this pipeline.

15. Conclusion

The documented system provides a reproducible, label-free approach to novelty detection in Mars HiRISE imagery. The V1 → V2 → V3 progression addresses reconstruction weaknesses through skip connections and then an explicit gradient-aware objective.

The final V3 encoder produces 256-D latent representations that are scored by a 300-tree Isolation Forest. The empirical threshold $\tau = 0.626565$ identifies 206 candidates (1.98%), with sample_06029.jpg ranked first at 0.742821.

Reconstruction-error localization and metadata comparison add interpretability and acquisition-context checks. The project appropriately frames geological interpretations as hypotheses rather than ground truth.

FINAL ANSWERS
Model: Autoencoder V3 with symmetric skip connections and gradient-aware loss
Novelty detector: 300-tree Isolation Forest on 256-D latent vectors
Threshold: $\tau = \mu + 3\sigma = 0.626565$
Candidates: 206 / 10,422 = 1.98%
Top-1: sample_06029.jpg — novelty score 0.742821
Top-5: sample_06029, sample_04537, sample_03547, sample_09253, sample_08233